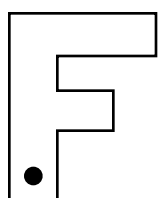
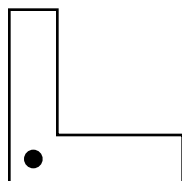
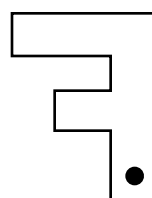


Only the tile outline counts. A face dot keeps track of which side is up.

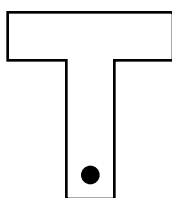
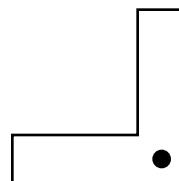
Problem 1: Use each flat tile and its mirror twin. Keep the dotted faces upward: which pairs can match by slides and turns? Then allow turning a tile over. Does the answer change?



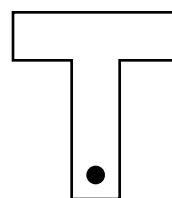
mirror



mirror

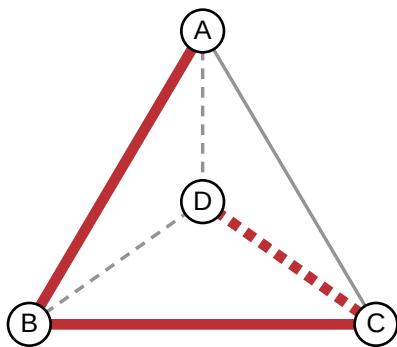


mirror

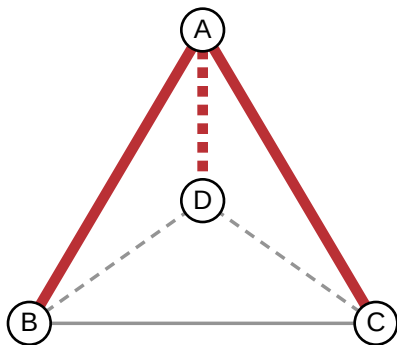
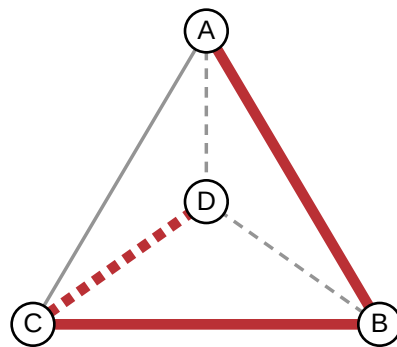


Use rigid regular tetrahedral frames. Letters locate edges while building; they do not count in a match. Red sleeves stay attached.

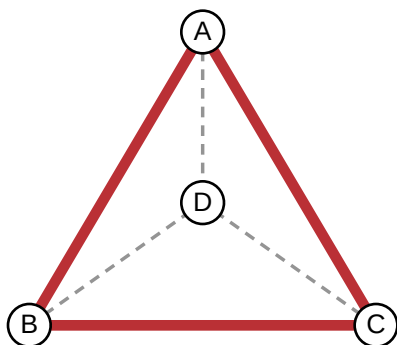
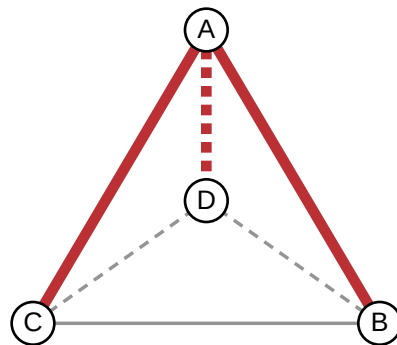
Problem 2: Build each edge pattern and its mirror. Which pairs can match by moving and turning? Swap one red sleeve to a plain edge to make a pair whose matching answer changes. Seek a reason that settles every possible turn.



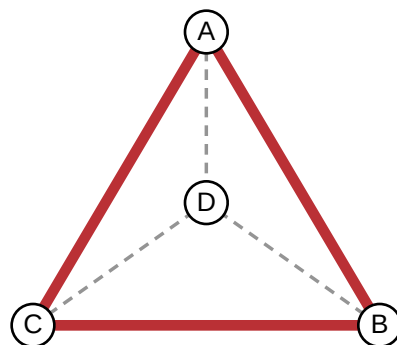
mirror



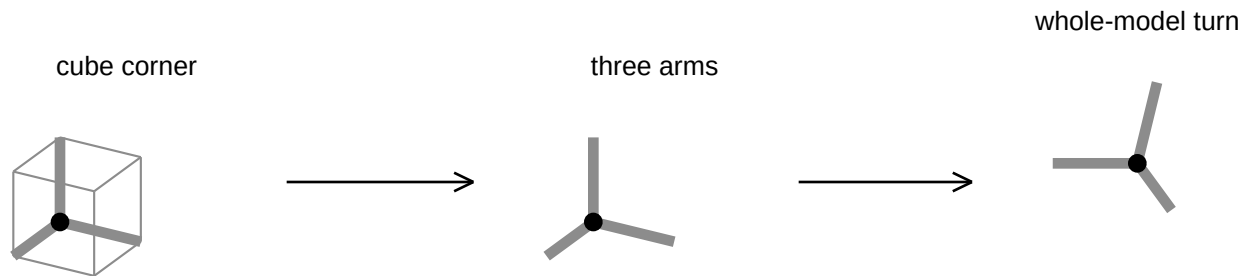
mirror



mirror



Each corner has three rigid arms meeting at right angles. Move and turn a whole model; keep its arms attached. Color names, not letter direction, count.



Problem 3: Build a corner and its mirror with three different arm colors, then with two alike, then with all alike. Which pairs can match by turning? Can unequal arm lengths make a mirror pair impossible to match when all arms have the same color?

